

案例 1.3 快速评卷的方案设计(英文)*

问题

在确定像数学建模竞赛这种形式比赛的优胜者时,常常要评阅大量的答卷,比如说,有 $P=100$ 份答卷,一个由 J 位评阅人组成的小组来完成评阅任务。基于竞赛资金,对于能够聘请的评阅人数量和评阅时间的限制,如果 $P=100$,通常取 $J=8$ 。理想的情况是每个评阅人看所有的答卷,并将它们一一排序,但这种方法工作量太大。另一种方法是进行一系列的筛选,在一次筛选中每个评阅人只看一定数量的答卷,并给出分数。为了减少所看答卷的数量,考虑如下的筛选模式:如果答卷是被排序的,则在每个评阅人给出的排序中排在最下面的 30% 答卷被剔除;如果答卷被

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打分(比如说从 1 分到 10 分),则某个截止分数线以下的答卷被筛选。这样,通过筛选的答卷重新放在一起返回给评阅小组,重复上述过程。人们关注的是,每个评阅人看的答卷总数要显著地小于 P ,评阅过程直到剩下 W 份答卷时停止,这些就是优胜者。当 $P=100$,通常取 $W=3$ 。

你的任务是利用排序、打分及其他方法的组合,确定一种筛选模式。按照这种模式,最后选中的 W 份答卷只能来自“最好的” $2W$ 份答卷(所谓“最好的”是指,我们假定存在着一种评阅人一致赞同的答卷的绝对排序)。例如,用你给出的方法得到的最后 3 份答卷将全部包括在“最好的”6 份答卷中。在所有满足上述要求的方法中,希望你能给出使每个评阅人所看答卷份数最少的一种方法。

注意在打分时存在系统偏差的可能。例如,对于一批答卷,一位评阅人平均给 70 分,而另一位可能给 80 分。在你给出的模式中如何调节尺度来适应竞赛参数(P , J 和 W)的变化?

Abstract

This article discusses selection schemes for J judges to select W winners from P papers. We develop two models, each of which first selects $2W$ papers. Then each judge gives the $2W$ papers numerical scores and selects W papers according to their average scores.

In Model 1, each judge reads part of the papers, gives them numerical scores, and picks out one top paper from those he has read, all these top papers forming a set A . Judge D_1 gives the papers of set A numerical scores and selects one top paper. After properly modifying set A , judge D_1 selects one top paper from set A . Repeat this process until $2W$ papers are selected. Improved

Model I overcomes the disadvantage that judge D_1 has to read too many papers.

In Model I, each numerical score is converted into standard score. Sequence all the papers in the order of their average standard scores, and the top $2W$ papers are picked out. We also give an improvement of Model I, where each judge's judging weight is gained by the principle of paired comparison matrix in Analytic Hierarchy Process.

Through the test of Model I' with 1000 simulations on PC computer, we get a reasonable result.

As to the data given in the problem, in Model I the average number of papers that each judge reads is 19.25, in Model I', 24; in the improvement of Model I, 28.

Assumptions

(1) By the "best" $2W$ papers, we mean there exists a set S_1 which consists of $2W$ papers. The top $2W$ papers in every judge's opinion are all in S_1 , i. e., if $a \in S_1, a' \notin S_1$, then each judge would give higher score to a than a' .

(2) Each judge gives scores by his own criterion, i. e., some judges tend to give higher scores, while others tend to give lower scores.

(3) Each judge gives the papers numerical scores.

(4) The same judge gives the same paper a unique score.

Restatement of the Problem

In a competition there are generally a large number of papers for judges to judge. A group of judges is collected to accomplish

the judging. Both the number of judges and amount of time are constrained. Our task is to determine a selection scheme, by a number of screening rounds, so that the total number of papers that each judge reads must be substantially less than the total number of the papers. Among the "best" $2W$ papers that all judges would agree, the final W papers will be selected from, which are the winners of the competition.

Symbols

- D_i the judge whose serial number is i
- a_i the paper whose serial number is i
- $m_{i,j}$ the numerical score of a_i given by judge D_j
- $M_{i,j}$ the average score of the group of papers P_j given by judge D_i
- $ms_{i,j}$ the standard score converted from $m_{i,j}$

Analysis of the Problem and Model Design

Model I

Based on Assumption 1, a convenient algorithm to select W winners is developed as following:

Algorithm 1

(1) Divide the P papers into J groups approximately equally. Each judge reads one group and gives them numerical scores. The top $2W$ papers of each group will be selected while the others will be rejected. Suppose the $2W$ papers selected by judge D_i are $a_{i,1}, a_{i,2}, \dots, a_{i,2W}$, ordered by scores descendingly.

(2) There is one top paper in every group. All these papers form a set A . That is, $A = \{a_{1,1}, a_{2,1}, \dots, a_{J,1}\}$. Each paper in

set A will be given a score by judge D_i .

(3) Suppose that the paper with the highest score in set A is $a_{i,k}$ (initially, $k=1$). It means that at step (1) this paper is read by Judge D_i , and ranks k th in the papers read by D_i . We prove that $a_{i,k} \in S_1$ (Details in Appendix).

(4) Record $a_{i,k}$, and replace $a_{i,k}$ with $a_{i,k+1}$ in set A . Judge D_i gives $a_{i,k+1}$ a score.

(5) Repeat step (3) ~ (4), until $2W$ papers of S_1 are found.

(6) Each judge gives these $2W$ papers numerical scores. According to the average scores of J scores, select the top W papers as winners.

For Algorithm 1, the average number of the papers each judge reads is $(P+J+2WJ-2)/J$. As to the data of this problem, it is 19.25.

Improvement of Model I

One obvious disadvantage of Model I is that judge D_i reads much more papers than the others, especially when J is large. So we improve Model I as following:

In order to describe the improved algorithm conveniently, we only demonstrate it with the data given in this problem.

Algorithm 2

(1) The same as step (1) of Algorithm 1.

(2) Divide the 8 top papers into two group A_1 and A_2 . That is, let

$$A_1 = \{a_{11}, a_{21}, a_{31}, a_{41}\} \quad A_2 = \{a_{51}, a_{61}, a_{71}, a_{81}\}$$

Also, let

$$B_1 = \{a_{12}, a_{22}, a_{32}, a_{42}\} \quad B_2 = \{a_{52}, a_{62}, a_{72}, a_{82}\}$$

(3) Judge D_1, D_3, D_5 and D_6 give scores to the papers in A_1 ,

A_2 , B_1 and B_2 respectively.

(4) Suppose the top paper in B_1 is a_{32} , so the corresponding paper in A_1 is a_{81} . Similarly, suppose the top paper in B_2 is a_{62} , and the corresponding paper in A_2 is a_{61} . Let $A'_1 = \{\text{papers belonging to } A_1 \text{ whose scores are no less than that of } a_{31}\}$. Suppose $A_1 = \{a_{11}, a_{31}, a_{41}\}$. Similarly, suppose $A'_2 = \{a_{61}, a_{81}\}$.

(5) Compare the top paper in A'_1 with that in A'_2 . That is, the two papers are handed to one judge (D_1 or D_5) and given scores. Suppose the better of them is a_{81} in A'_2 . We can prove that $a_{81} \in S_1$ in a similar way. Eliminate a_{81} from A'_2 and A_2 , and repeat step (5). Suppose after 3 times of comparison, 3 papers a_{81} , a_{11} and a_{61} are found to belong to S_1 .

(6) Now A'_2 is empty. For the eliminated papers in A_2 , move the corresponding papers in B_2 from B_2 to A_2 . For the removed papers in B_2 , the process is similar. That is, let $A_2 = \{a_{31}, a_{62}, a_{71}, a_{82}\}$, $B_2 = \{a_{52}, a_{63}, a_{72}, a_{83}\}$.

(7) Repeat step (3) ~ (6) until $2W=6$ papers belonging to S_1 are found.

(8) The same as step (6) of Algorithm 1.

Improved Model I slightly increases the total work load, but it makes the work load of each judge more average and reduces the work time. It overcomes the disadvantage of Model I to a certain extent.

When J is large, the J top papers selected in step (1) can be divided into more than 2 groups, to make the work load of each judge more average.

Model I

The inconsistency of each judge's own criterion prevents us

from judging in the same standard. However, once the criterion of judge $D_i (1 \leq i \leq J)$ is specified as the standard, scores given by all the other judges can be in a certain way converted into the standard scores based on the criterion of judge D_i (Abbreviated as the standard scores to judge D_i). We call the conversion method Conversion of Standard Scores. The method is shown as following:

J judges are seated clockwise. Every two adjacent judges have read the same group of papers. We assume that judge D_i and his left-hand side judge D_j both read the same group of papers named P_1 . The average scores are $M_{1,i}$ and $M_{1,j}$ respectively. We conclude that the criterion of judge D_j to that of judge D_i is $M_{1,i}/M_{1,j}$. We call $M_{1,i}/M_{1,j}$ the conversion ratio $\text{con}(i, j)$. When converting the scores of judge D_j into the standard scores to judge D_i , we multiply these scores by $M_{1,i}/M_{1,j}$. The standard scores of judge D_j to his right-hand side judge D_k can be obtained similarly.

There is a relationship between the criteria of the two adjacent judges. Once the criterion of a certain judge (say, D_a) is specified as the standard, criteria of every judges (D_b) can all be converted successively with that of the judge next to him. In the end we get the conversion ratio $\text{con}(b, a)$ of D_b and D_a . By this means, scores of every judge are converted into standard scores.

Naturally the consistence of J conversion ratio should be under consideration. If

$$\text{con}(2, 1)\text{con}(3, 2)\cdots\text{con}(J, J-1)\text{con}(1, J) = \beta \neq 1$$

then use the division of each conversion ratio by $\beta^{\frac{1}{J}}$ as the final conversion ratio.

Model I is on the basis of the above idea. All the judges read a certain number of papers. After score selection scheme the papers are sequenced in the order of standard score. In this algorithm we use the criterion of D_1 as the standard. The conversion ratio of judge D_i to D_1 , when $2 \leq i \leq [J/2] + 1$, is converted in the order of $D_i, D_{i-1}, \dots, D_1$; the reverse, in the order of $D_i, D_{i+1}, \dots, D_J, D_1$. The principle is to make the least number of conversion.

The algorithm is shown as following:

(1) Divide P papers into J groups: $P_1, \dots, P_J$ approximately equally. Each judge reads a group of papers.

(2) When the judging is accomplished, in each group $2W$ papers which get higher scores are left and the others are rejected. Suppose the new groups are $P'_1, \dots, P'_J$.

(3) Judges exchange their groups clockwise, i. e., judge D_i gives group P'_i to his right-hand side judge ($i=1, \dots, J$).

(4) In this way each group is read by two judges. According to the conversion method of standard scores, we get the conversion ratio $\text{con}(j,1)$ of judge D_j ($j=2, 3, \dots, J$) to D_1 .

(5) Through the above steps, each paper a_k gets two standard scores $ms_{k,i}$ and $ms_{k,j}$ respectively from judge D_i and judge D_j . The final standard score is $(ms_{k,i} + ms_{k,j})/2$.

(6) Each selected paper a_k of $P'_1, \dots, P'_J$ get an average standard score. Sequence these $2WJ$ papers in the order of standard scores and the top $2W$ papers are then selected.

(7) In order to select W papers among the $2W$, every judge must read all the $2W$ papers. Score of every paper is converted into standard score.

(8) Each one of the $2W$ papers gets J standard scores. Sequence the papers in the order of the average scores of their standard scores. The top W papers are the winners.

Conversion method of standard score can also be applied to the papers rejected in the first round. In deed, each one of the P papers can get a standard score, so all papers can be sequenced in the order of standard scores.

In this algorithm, the average number of papers read by each judge is $(P + 4WJ - 4W) / J$. The number is 23 as to the data given in the problem.

Further Discussion of Model I

In Model I, we get a group of direct conversion ratio of the criterions of every two adjacent judges. It inspires us that if we can get the direct conversion ratio of the criterions of every two judges, then each judge's judging weight will be gained by the principle of paired comparison matrix in Analytic Hierarchy Process.

We assume every two judges have read a group of identical papers, therefore paired comparison matrix X is constituted. $J(J+1)/2$ data which are required in X can be gained by a method explained in the ensuring chapters of paragraphs. Though it could not ensure that X is a consistent order matrix, the inconsistent level is in a permitted scope, according to assumption 2. On the basis of the principle about paired comparison matrix, the eigenvector of the eigenvalue of maximum λ of X is regarded as weight vector ω , let

$$\omega = (\omega_1, \omega_2, \dots, \omega_J)$$

then each element ω_i ($i=1, \dots, J$) can be regarded as the judging

weight of judge D_i .

Two key problems are to be resolved. One is how to get the data in X and the other is how to get λ and ω . The latter will be obtained by radical method, a practical algorithm. To the former, we give a solution when $J=8$, $2W=6$ and $P=100$. Arrange the exchanging papers with a certain method. The arrangement is shown as Tab. 1.3.1:

Tab. 1.3.1

Group Judge Description	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8
	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8
Round One	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
Round Two	P_2'	P_3'	P_4'	P_5'	P_6'	P_7'	P_8'	P_1'
Round Three	P_4'	P_5'	P_6'	P_7'	P_8'	P_1'	P_2'	P_3'

Every group of papers is read by three judges, so scores given to this group of papers can determine the direct conversion ratio of the criterions of every two of these three judges. 8 groups of papers can determine 24 direct conversion ratio at the most. Once paper exchanges are arranged properly, these 24 ratio will not repeat themselves. As shown in Fig. 1.3.1, the value of $a_{i,j}$ which is marked with "O" has been determined ($a_{i,j}=1/a_{j,i}$).

Fig. 1.3.1 shows that a_{15} , a_{26} , a_{37} and a_{48} haven't been determined yet. Of course, if we exchange groups of papers again and continue to read papers, these four values will be determined through direct comparison, but it will result in each judge's reading 6 papers more. We consider it is not worth increasing 25% of

work load to each judge in order to get accuracy of 11% of data. It is more reasonable to get these 4 data indirectly on the basis of other data.

$\begin{matrix} j \\ x_{i,j} \\ i \end{matrix}$	1	2	3	4	5	6	7	8
1	○	○	○	○		○	○	○
2		○	○	○	○		○	○
3			○	○	○	○		○
4				○	○	○	○	
5					○	○	○	○
6						○	○	○
7							○	○
8								○

Fig. 1. 3. 1

In the improved model, the average number of papers each judge reads is 28 as to the data given by this problem.

Test of Model I

We assume a paper can be really evaluated by a numerical score between 0 to 100, which is called absolute score.

When the absolute score of a certain paper is m , we assume the score b_i that judge D_i gives to the paper is $k_i m + e_i$, where k_i called systematic bias determines the criterion of judge D_i and k_i follows normal distribution $N(1, \sigma_0^2)$ (in practice, $\sigma_0 = 0.02$, and the deviation will be controlled within 10); e_i is random error, which follows normal distribution $N(0, \sigma_i^2)$, and random variable σ_i determines the fluctuation of e_i , which follows normal distribu-

tion $N(\mu, \sigma_0^2)$ (in practice, let $\mu=1$ and $\sigma_0=0.5$, then the fluctuation of e_i will be controlled within 5).

Therefore we can use a computer to create absolute scores of P papers randomly (we assume these scores follow normal distribution $N(60, 17)$), systematic bias k_i of each judge, and deviation of random error σ_i . According to the algorithm of Model I, find the top 6 rank and count how many papers of them belong to the top 6 papers ranked by absolute score to test the model.

Making simulations 1000 times, we know that, on average there are 5 papers belonging to the top 6 papers ranked by absolute score. Thus Model I proves reasonable.

Strengths and Weaknesses & Evaluation of Model

In Model I, each judge reads a less number of papers while the work load goes to a certain judge. As a result the work time is prolonged. In improved Model I, work is approximately assigned equally to each judge and the cost is the increasing number of papers that each judge has to read.

In Model I, all the P papers are ranked for the judges to select between different classes of papers. Model I synthesizes the opinions of the judges and effectively eliminate the affection of systematic bias. On the other hand, there is a certain error since the conversion ratio of the criterion of each judge D_i ($2 \leq i \leq J$) to that of judge D_1 is not worked out directly but is obtained through indirect conversion. In the improved Model I we get the judging weight of each judge. Error rate is reduced while work

load of each judge increases.

Model I is used when total work load and the number of judges J are relatively less. Model II is used when the symmetry of each judge's work load is required. When there are enough funding and time and low error rate is required, we recommend Model II and its improvement.

Appendix

Theorem 1

Before $2W$ papers belonging to S_1 are found, the top paper a_{ii} in set A selected by judge D_i belongs to S_1 .

We prove a lemma before proving theorem 1:

Lemma

Suppose a judge gives papers a and a' numerical scores, and the score of a is higher than that of a' . We can draw the conclusion that if $a \notin S_1$, $a' \notin S_1$.

Proof If $a \notin S_1$, but $a' \notin S_1$, according to Assumption 1, the score of a' is higher than a . This is contrary to the condition of the lemma. Thus the lemma is proved.

Proof of Theorem 1 Arrange the $2WJ$ papers selected in step (1) as a matrix;

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